

Supplemental Online Material - Tables of Measurements

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*Measurement of the EMC Effect of Tritium and Helium-3 by the JLab
MARATHON Experiment*

x	Q^2 (GeV/ c) ²	W GeV/ c^2	σ_h/σ_d	Δ_{stat}	Δ_{ptp}	Δ_{syst}	Δ_{tot}
0.195	2.73	3.49	1.071	0.003	0.005	0.005	0.008
0.225	3.15	3.42	1.071	0.003	0.005	0.005	0.008
0.255	3.57	3.36	1.071	0.004	0.005	0.005	0.008
0.285	3.99	3.30	1.078	0.004	0.005	0.005	0.009
0.315	4.41	3.24	1.064	0.005	0.005	0.005	0.009
0.345	4.83	3.17	1.078	0.005	0.005	0.005	0.009
0.375	5.25	3.10	1.089	0.007	0.005	0.005	0.011
0.405	5.67	3.03	1.079	0.009	0.005	0.005	0.012
0.435	6.09	2.96	1.064	0.009	0.005	0.005	0.012
0.465	6.51	2.89	1.081	0.009	0.005	0.005	0.012
0.495	6.93	2.82	1.102	0.009	0.005	0.006	0.012
0.525	7.35	2.74	1.106	0.010	0.005	0.006	0.013
0.555	7.77	2.67	1.089	0.010	0.005	0.006	0.013
0.585	8.19	2.59	1.083	0.010	0.005	0.006	0.013
0.615	8.61	2.50	1.082	0.010	0.005	0.006	0.013
0.645	9.03	2.42	1.106	0.011	0.005	0.006	0.014
0.675	9.45	2.33	1.079	0.011	0.005	0.006	0.014
0.705	9.87	2.24	1.117	0.011	0.005	0.006	0.014
0.735	10.3	2.14	1.118	0.011	0.005	0.006	0.014
0.765	10.7	2.04	1.128	0.011	0.005	0.006	0.014
0.795	11.1	1.94	1.130	0.013	0.005	0.006	0.015
0.825	11.9	1.84	1.148	0.016	0.006	0.006	0.018

Table 1: The helion (h , ^3He) to deuteron (d , ^2H) DIS cross section ratio (see text) for the x , Q^2 , and W kinematics of the MARATHON JLab Experiment. At the highest central- x kinematics of 0.825 only electron events with $W > 1.74$ GeV/ c^2 were included in the determination of the helion and deuteron yields. Listed also are the statistical (stat), point-to-point (ptp), and overall normalization systematic (syst) components of the total (tot) error. The latter is the quadrature of the three components.

x	Q^2 (GeV/ c) ²	W GeV/ c^2	σ_t/σ_d	Δ_{stat}	Δ_{ptp}	Δ_{syst}	Δ_{tot}
0.195	2.73	3.49	0.962	0.002	0.005	0.005	0.008
0.225	3.15	3.42	0.952	0.002	0.005	0.005	0.008
0.255	3.57	3.36	0.937	0.003	0.005	0.005	0.008
0.285	3.99	3.30	0.928	0.004	0.005	0.005	0.008
0.315	4.41	3.24	0.920	0.004	0.005	0.005	0.008
0.345	4.83	3.17	0.920	0.005	0.005	0.005	0.009
0.375	5.25	3.10	0.923	0.006	0.005	0.005	0.010
0.405	5.67	3.03	0.884	0.007	0.005	0.005	0.010
0.435	6.09	2.96	0.880	0.007	0.005	0.005	0.010
0.465	6.51	2.89	0.880	0.008	0.005	0.005	0.010
0.495	6.93	2.82	0.890	0.007	0.005	0.005	0.010
0.525	7.35	2.74	0.876	0.008	0.005	0.005	0.010
0.555	7.77	2.67	0.858	0.008	0.005	0.005	0.010
0.585	8.19	2.59	0.852	0.008	0.005	0.005	0.010
0.615	8.61	2.50	0.851	0.008	0.005	0.005	0.013
0.645	9.03	2.42	0.854	0.009	0.005	0.005	0.011
0.675	9.45	2.33	0.835	0.009	0.005	0.005	0.011
0.705	9.87	2.24	0.851	0.099	0.005	0.005	0.011
0.735	10.3	2.14	0.856	0.009	0.005	0.005	0.011
0.765	10.7	2.04	0.855	0.009	0.005	0.005	0.011
0.795	11.1	1.94	0.859	0.010	0.005	0.005	0.012
0.825	11.9	1.84	0.880	0.013	0.005	0.005	0.014

Table 2: The triton (t , ${}^3\text{H}$) to deuteron DIS cross section ratio (see text) for the x , Q^2 , and W kinematics of the MARATHON JLab Experiment. At the highest central- x kinematics of 0.825 only electron events with $W > 1.74$ GeV/ c^2 were included in the determination of the triton and deuteron yields. Listed also are the statistical (stat), point-to-point (ptp), and overall normalization systematic (syst) components of the total (tot) error. The latter is the quadrature of the three components.

x	Q^2 (GeV/ c) ²	W GeV/ c^2	$[(\sigma_t + \sigma_h)/(2\sigma_d)]$	Δ_{stat}	$\Delta_{\text{stat+ptp}}$	Δ_{tot}
0.195	2.73	3.49	1.016	0.002	0.006	0.010
0.225	3.15	3.42	1.011	0.003	0.006	0.009
0.255	3.57	3.36	1.004	0.003	0.006	0.010
0.285	3.99	3.30	1.003	0.004	0.006	0.010
0.315	4.41	3.24	0.992	0.004	0.007	0.010
0.345	4.83	3.17	0.999	0.005	0.007	0.011
0.375	5.25	3.10	1.006	0.006	0.008	0.011
0.405	5.67	3.03	0.981	0.006	0.008	0.011
0.435	6.09	2.96	0.972	0.007	0.009	0.011
0.465	6.51	2.89	0.980	0.007	0.009	0.012
0.495	6.93	2.82	0.996	0.008	0.010	0.012
0.525	7.35	2.74	0.991	0.008	0.010	0.012
0.555	7.77	2.67	0.974	0.008	0.010	0.012
0.585	8.19	2.59	0.967	0.008	0.009	0.012
0.615	8.61	2.50	0.967	0.008	0.009	0.012
0.645	9.03	2.42	0.980	0.008	0.009	0.012
0.675	9.45	2.33	0.957	0.008	0.009	0.012
0.705	9.87	2.24	0.984	0.009	0.010	0.012
0.735	10.3	2.14	0.987	0.009	0.010	0.013
0.765	10.7	2.04	0.992	0.009	0.011	0.013
0.795	11.1	1.94	0.995	0.010	0.012	0.014
0.825	11.9	1.84	1.014	0.011	0.014	0.016

Table 3: The EMC effect ratio of the “ $A = 3$ nucleus” $[(\sigma_t + \sigma_h)/(2\sigma_d)]$ DIS cross section ratio for the x , Q^2 , and W kinematics of MARATHON JLab Experiment. Listed also are the statistical (stat), and combined statistical and point-to-point (stat+ptp) components of the also listed total (tot) error. The latter is the quadrature of the statistical, point-to-point, and overall systematic normalization components.

x	Q^2 (GeV/ c) ²	F_{iso}	$(\sigma_h/\sigma_d)_{\text{iso}}$	Δ_{stat}	$\Delta_{\text{stat+ptp}}$	Δ_{tot}
0.195	2.73	0.950	1.016	0.002	0.006	0.010
0.225	3.15	0.944	1.011	0.003	0.006	0.009
0.255	3.57	0.937	1.003	0.003	0.006	0.010
0.285	3.99	0.930	1.003	0.004	0.006	0.010
0.315	4.41	0.932	0.992	0.004	0.007	0.010
0.345	4.83	0.927	0.999	0.005	0.007	0.011
0.375	5.25	0.924	1.007	0.006	0.008	0.011
0.405	5.67	0.910	0.982	0.006	0.008	0.011
0.435	6.09	0.915	0.974	0.007	0.009	0.011
0.465	6.51	0.909	0.982	0.007	0.009	0.012
0.495	6.93	0.906	0.998	0.008	0.010	0.012
0.525	7.35	0.898	0.993	0.008	0.010	0.012
0.555	7.77	0.896	0.977	0.008	0.010	0.012
0.585	8.19	0.896	0.970	0.008	0.009	0.012
0.615	8.61	0.896	0.970	0.008	0.009	0.012
0.645	9.03	0.889	0.984	0.008	0.009	0.012
0.675	9.45	0.891	0.961	0.008	0.009	0.012
0.705	9.87	0.885	0.989	0.009	0.010	0.012
0.735	10.3	0.887	0.992	0.009	0.010	0.013
0.765	10.7	0.883	0.996	0.009	0.011	0.013
0.795	11.1	0.884	1.000	0.010	0.012	0.014
0.825	11.9	0.888	1.020	0.011	0.014	0.016

Table 4: The isoscalar EMC effect ratio for the ${}^3\text{He}$ (h) nucleus $(\sigma_h/\sigma_d)_{\text{iso}}$ DIS cross section ratio along with the associated isoscalarity correction factor F_{iso} (see text) at the x , Q^2 , and W kinematics of the MARATHON JLab Experiment. Listed also are the statistical (stat), and combined statistical and point-to-point (stat+ptp) components of the also listed total (tot) error. The latter is the quadrature of the statistical, point-to-point, and overall systematic normalization components.

x	Q^2 (GeV/c) ²	F_{iso}	$(\sigma_t/\sigma_d)_{\text{iso}}$	Δ_{stat}	$\Delta_{\text{stat+ptp}}$	Δ_{tot}
0.195	2.73	1.057	1.017	0.002	0.006	0.010
0.225	3.15	1.063	1.012	0.003	0.006	0.009
0.255	3.57	1.072	1.004	0.003	0.006	0.010
0.285	3.99	1.081	1.003	0.004	0.006	0.010
0.315	4.41	1.078	0.992	0.004	0.007	0.010
0.345	4.83	1.086	0.999	0.005	0.007	0.011
0.375	5.25	1.090	1.006	0.006	0.008	0.011
0.405	5.67	1.109	0.980	0.006	0.008	0.011
0.435	6.09	1.103	0.971	0.007	0.009	0.011
0.465	6.51	1.102	0.978	0.007	0.009	0.012
0.495	6.93	1.116	0.993	0.008	0.009	0.012
0.525	7.35	1.128	0.988	0.008	0.010	0.012
0.555	7.77	1.131	0.970	0.008	0.009	0.012
0.585	8.19	1.131	0.963	0.008	0.009	0.012
0.615	8.61	1.139	0.962	0.008	0.009	0.012
0.645	9.03	1.143	0.976	0.008	0.009	0.012
0.675	9.45	1.140	0.952	0.008	0.009	0.012
0.705	9.87	1.150	0.979	0.009	0.010	0.012
0.735	10.3	1.146	0.981	0.009	0.010	0.012
0.765	10.7	1.152	0.985	0.009	0.011	0.013
0.795	11.1	1.150	0.988	0.010	0.012	0.014
0.825	11.9	1.144	1.007	0.011	0.013	0.016

Table 5: The isoscalar EMC effect ratio for the ^3H (h) nucleus $(\sigma_t/\sigma_d)_{\text{iso}}$ DIS cross section ratio along with the associated isoscalarity correction factor F_{iso} (see text) at the x , Q^2 , and W kinematics of the MARATHON JLab Experiment. Listed also are the statistical (stat), and combined statistical and point-to-point (stat+ptp) components of the also listed total (tot) error. The latter is the quadrature of the statistical, point-to-point, and overall systematic normalization components.